

Problem | Test Cases | Output |

Explanation:

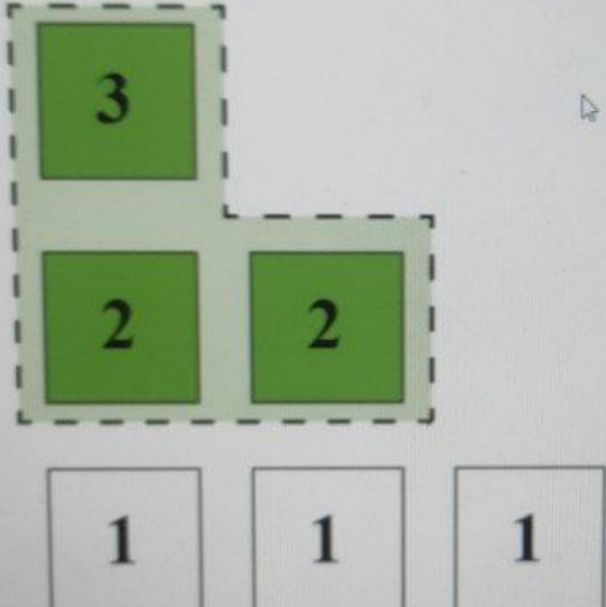
Product 0
1 box with 3 units

Product 1
2 boxes with 2 units each

Product 2
3 boxes with 1 unit each

The maximum number of units that can be shipped

$= 3 + 2 + 2 = 7$ units



Razorpay

1. Popular Shops


```

        pref[i+1][c - 'a']++;
    }
}
int q;
cin >> q;
while(q--){
    int l,r,k;
    cin >> l >> r >> k,l--;
    int cnt = 0;
    for(int i = 0; i < 26; i++){
        cnt += pref[r][i] - pref[l][i];
        if(cnt >= k){
            cout << (char) ('a' + i) << '\n';
            break;
        }
    }
}
}
}
}

```

variables "mutex_a" and

is the output of the following C

ta structures, what is the

ge case complexity analysis for

ing questions

ing traps

cter at K

shops

10

4.0

10

4.0

10

50.0

50.0

50.0

Question 17

Max score: 50.00

Character at K

You are given a list that contains N strings of lowercase English alphabets. Any number of contiguous strings can be grouped together to form a new string. The grouping function accepts two integers X and Y and concatenates all strings between indices X and Y (inclusive) and returns a modified string in which the alphabets of the concatenated string are sorted.

You are asked Q questions each containing two integers L and R . Determine the K^{th} character in the concatenated string if we pass L and R to the grouping function.

Input format

- First line: N (number of strings in the list)
- Next N lines: String S_i
- Next line: Q (number of questions)
- Next Q lines: Three space-separated integers, L , R , and K

Output format

13. If there are two processes 'A' and 'B' and variables 'mutex_a' and + 4.0
- 1.0

14. What is the output of the following C code. + 4.0
- 1.0

15. In Data structures, what is the average case complexity analysis for + 4.0
- 1.0

3 Programming questions

16. Avoiding traps + 50.0

17. Character at K + 50.0

18. Popular shops + 50.0

• Next Q lines: Three space-separated integers, L , R , and K

Output format

For each question, print the K^{th} character of the concatenated string in a new line.

Constraints

$1 \leq N \leq 10^5$

$1 \leq \text{len}(S_i) \leq 50$

$1 \leq Q \leq 10^5$

$1 \leq L \leq R \leq N$

It is always guaranteed that the K^{th} position is valid.

Sample input 1

```
5
aaaaa
bbbbb
```

Sample output 1

```
c
d
e
```

4. Avoiding Traps

of the following C + 4.0
- 1.0

what is the complexity analysis for + 4.0
- 1.0

+ 50.0

+ 50.0

+ 50.0

Question 16

Avoiding traps

There is a cave of N cells where each cell has a trap or is safe to land. From a cell i , you can jump to cells $i + 1$ or $i + 2$. Also, if number i is special, then you can also jump from cell i to cell $i + A$ where $A = \text{number of primes in } [1, i]$. A number i can be special if $\frac{A}{i} \geq \frac{r_1}{r_2}$.

You are given the following:

- Some arbitrary numbers r_1 and r_2
- The number of cells N in the cave where each cell contains either '#' or '*'
- Details of the cave

Note: Here, '#' represents an empty cell and '*' means a trapped cell.

Your task is to determine the minimum number of steps that you have to walk to reach the N^{th} cell. Initially, you are at cell 1.

Input format

There are two processes 'A' and 'B' with variables 'mutex_a' and 'mutex_b'. The output of the following C code is:

For each test case, print the minimum number of steps that you have to walk to reach the N^{th} cell. Initially, you are at cell 1.

Input format

- The first line contains an integer T denoting the number of test cases.
- The first line of each test case contains two integers r_1 and r_2 .
- The second line of each test case contains an integer N .
- The third line contains a string of length N representing N cells where the first character of the string represents the first cell, second character represents the second cell, and so on. Each cell is either '#' or '*'.

Output format

For each test case, print the minimum number of steps that you have to walk to reach the N^{th} cell. Print the output in a separate line.

If it is not possible to reach the N^{th} cell, then print "No way!" without quotes.

Constraints

$1 < T < 500$

5. Adjacent Cubes

<https://www.geeksforgeeks.org/minimize-the-maximum-difference-between-adjacent-elements-in-a-n-array/>

00:21:42 shashank2001@nitg.ac.in End test

Question 16 Max. score: 50.00

Adjacent cubes

You are given N cubes of wood arranged in **strictly non-decreasing order**. The length of the side of the i^{th} cube is $x[i]$ and side lengths of all the cubes are distinct. The **first** and the **last** cubes are glued at their positions and they cannot be moved.

The difference, $diff$, of the cubes is defined as the **maximum** difference of the sides of two adjacent cubes for all the cubes in the arrangement

$$diff = \max(x[i+1] - x[i]) \text{ for } 1 \leq i \leq N-1$$

You are required to **minimize** the $diff$ value of the cube arrangement by removing exactly K cubes.

Input format

- The first line contains a single integer N .
- The second line contains N space-separated integers $x[i]$ where $x[i]$ is the length of the side of the i^{th} cube.
- The third line contains a single integer K .

Output format

Print a single integer that is the **minimum** $diff$ value for the cube arrangement after removing K cubes.

Constraints

Guwahati In... 00:21:24 shashank2001@itg.ac.in End test

g uses the divide
e? +4.0 -1.0

g is/are incorrect?
needed. +4.0 -1.0

where a long
ask has acquired +4.0 -1.0

encing, the
to or more load +4.0 -1.0

+50.0

+50.0

+50.0

Output format

Print a single integer that is the **minimum diff** value for the cube arrangement after removing K cubes.

Constraints

$3 \leq N \leq 10^5$

$1 \leq K \leq N - 2$

$1 \leq x[i] \leq 10^9$

Sample input 1 Copy

```
5
1 2 3 7 8
2
```

Sample output 1 Copy

```
5
```

Explanation

In 3 ways we can remove K elements

The 3 cases are (1,7,8) (1,3,8) and (1,2,8). The awkwardness in the three cases are :

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6.Ways to generate

Question 18 Max. score

Ways to Generate

Given a string S of length N . And we have a set D , which has K unique strings. The sum of lengths of all the strings in set D is M .

Find the number of distinct ways string S can be generated as concatenation of strings in set D .

Each string in D can be used any number of times.

Since the value can be large. Print it modulo $10^9 + 9$.

Input Format

First line contains 2 space separated integers N and M

Second Line contains string S

Third line contains single Integer K representing number of strings in set D

Next K lines contain strings in set D

Constraints

$1 \leq N, M, K \leq 10^5$

All characters in all strings fall in range $[A, Z]$

+4.0 -1.0

+4.0 -1.0

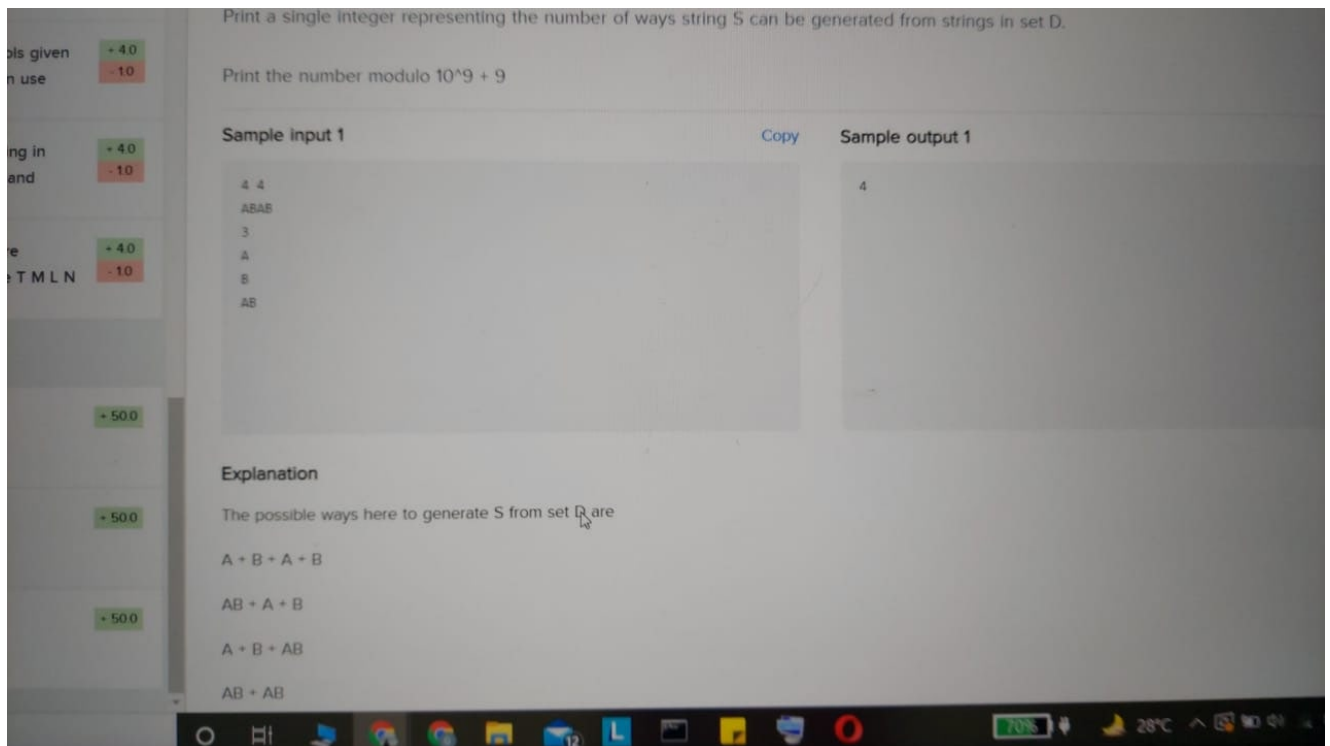
+4.0 -1.0

+4.0 -1.0

+50.0

+50.0

+50.0



7. Butterfly affect

(For every test case print the number of arrangements mod)

<https://www.hackerearth.com/problem/algorithm/butterfly-effect-4e5d6946/>

8. <https://leetcode.com/discuss/interview-question/906248/Interview-Question/768089>

There are N stones lying in a line. The cost and type of the ith stone is 'ai' units and 'i' respectively. You are initially having zero stones and you wish to collect all the N types of stones, type 1, type 2,.....,type N.

9. <https://www.geeksforgeeks.org/find-two-numbers-from-their-sum-and-xor-set-2/>

VISA